Charged particles decrease in velocity as they lose their energy in traversing an absorber. As a result they spend progressively longer times in the vicinity of any particular atom, which results in an increase in the probability of interaction with that atom. Consequently there is a steady increase in the number of ion pairs formed along the path (or track) of the particle rather than a constant density of ion pairs. As mentioned above, near the end of the range for heavy charged particles a maximum is observed for the number of ion pairs formed per unit path length (the *Bragg peak*) (Fig. 7.7b). At a distance just beyond the Bragg peak maximum the kinetic energy of the particles is comparable to those of the orbital electrons of the absorber. As a result, the particle can acquire electrons, $\gamma < 1$ in Eqn. (7.14), finally becoming uncharged, $\gamma = 0$, and thereby losing its ability to cause further ionization. When scaled by the maximum ionization, the Bragg peak is most pronounced for protons.

7.4. ABSORPTION OF ELECTRONS

Absorption of high energy electron beams occurs through interaction with the orbital electrons and the electromagnetic field at the atom. The processes are summarized in Table 7.1 and Figure 7.8. In order to distinguish between electrons from accelerators and those from β-decay we refer to the latter as β-*particles*.

7.4.1. Ionization

Beta-particles lose their energy primarily by the same mechanism as α-particles (Fig. 7.7a and b); however, there are several important differences. Since the masses of the β-particles and of the orbital electrons are the same at non-relativistic velocities, the β-particles can lose a large fraction of their energy in a single collision. The β-particle undergoes a wide angle deflection in such collisions and consequently β-particles are scattered out of the beam path all along the length. The secondary electrons ionized from the atom have such high energies that they cause extensive secondary ionization which provides 70—80% of the total ionization in β-absorption processes (Fig. 7.5b). Approximately half of the total energy of the β-particle is lost by ionization and half by excitation. The track formed is further discussed in §8.2.

The specific ionization from a β-particle is much lower than that from a heavy ion as can be seen in Figure 7.7a. This is due to the fact that for the same initial energy, β-particles have much greater velocity than have α-particles or protons because their mass is very much smaller than the mass of the heavy particles. This greater velocity results in a correspondingly lower ionization due to the considerable shorter time available for interaction with the electrons of a specific atom and gives a much longer range to β-particles. The erratic path observed for β-particles in Figure 7.5(c) is a result of the large energy transfer and the resulting large deflections involved in the encounters with the orbital electrons. However, at very high energies the β-particles have straight

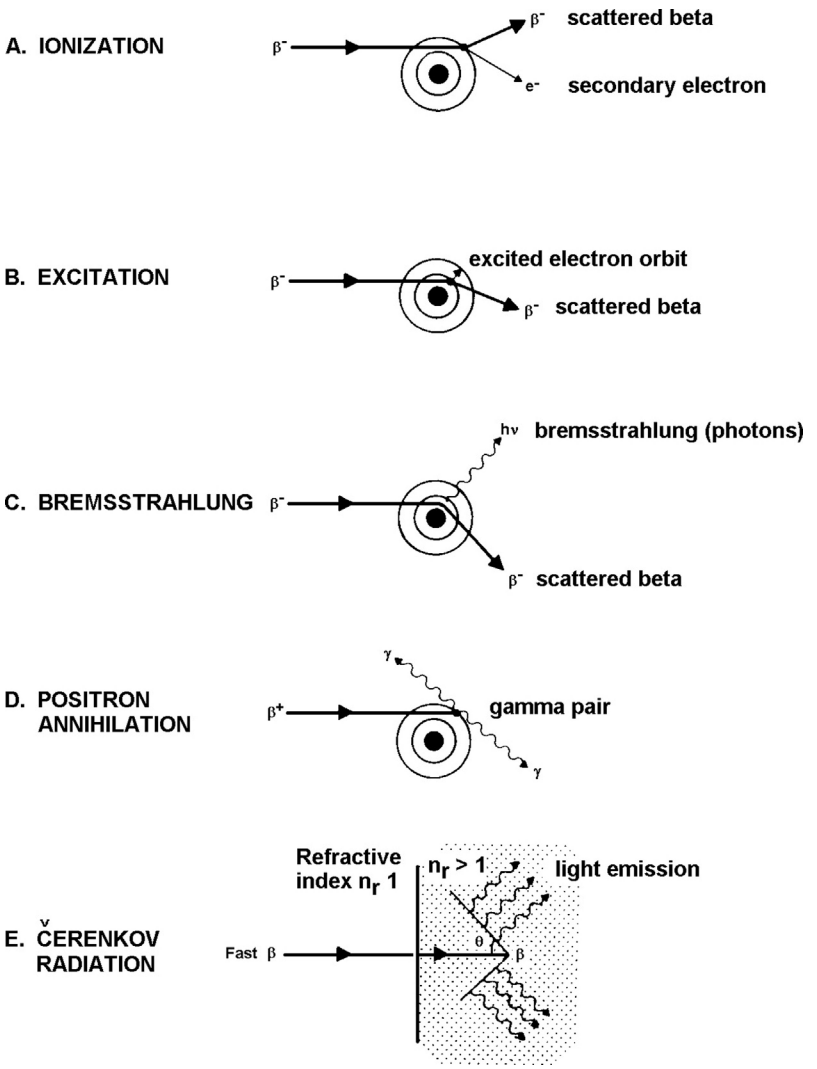


Figure 7.8 Schematic description of the five processes accounting for β-particle absorption.

paths as a result of the fact that very energetic β-particles have a momentum considerably in excess of that of the orbital electron.

7.4.2. Bremsstrahlung

As a β-particle approaches an atomic nucleus, it is attracted by the positive field of the nucleus and deflected from its path. The deflection results in an acceleration that, according to classical electrodynamics, leads to emission of electromagnetic radiation (Fig. 7.8c), cf. synchrotron radiation Ch. 16. Therefore the encounter with the positive

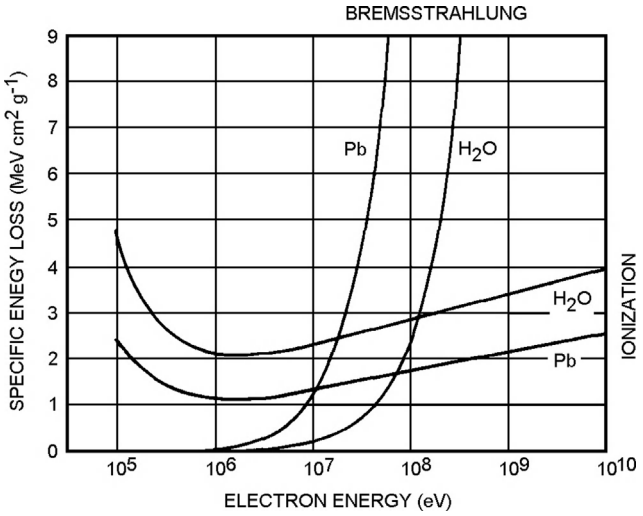


Figure 7.9 Energy loss of fast electrons by ionization and bremsstrahlung. *(From Gentner, Maier-Leibnitz, and Bothe.)*

charge of the nuclear field decreases the energy of the β-particle by an amount exactly equal to the amount of electromagnetic radiation emitted. This radiation is known as *bremsstrahlung* (braking radiation). The loss of energy by emission of bremsstrahlung radiation increases with the β energy and with the atomic number (positive charge) of the absorber material (Fig. 7.9). In aluminum approximately 1% of the energy of a 1 MeV electron is lost by bremsstrahlung radiation and 99% by ionization whereas in lead the loss by bremsstrahlung is about 10%. For electrons of greater than 10 MeV energy, bremsstrahlung emission is the predominant mode of energy loss in lead. However, for the energies in radioactive decay, bremsstrahlung can usually be neglected - particularly for absorption processes in material of low atomic weight. The ratio of *specific energy loss* (dE/dx) through bremsstrahlung to that through collision (i.e. all other processes) is approximately:

$$(dE/dx)_{\text{brems}}/(dE/dx)_{\text{coll}} \approx E_e Z/800 \tag{7.16}$$

where E_e is kinetic energy of the electron (MeV) and Z the atomic number of the absorber atoms.

Figure 7.10 shows the bremsstrahlung spectrum obtained in aluminum for β-particles emitted by ^{147}Pm. In this case a very small fraction of the β-energy is converted into radiation. The bremsstrahlung spectrum is always of much lower energy than that of the β-spectrum. Bremsstrahlung sources of a wide variety of energies are commercially available. Recently special electron accelerators have been designed for the sole purpose of producing bremsstrahlung radiation to be used for (analytical) X-ray excitation and for medical irradiation purposes; see Ch. 16.

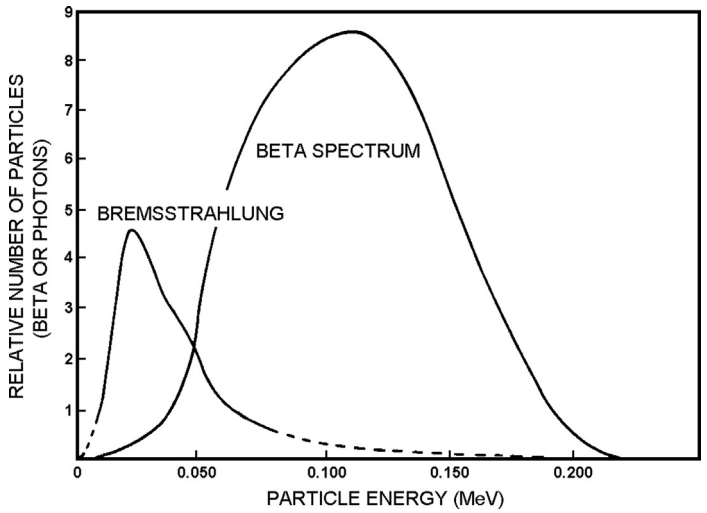


Figure 7.10 Beta-spectrum (right curve) and bremsstrahlung spectrum in aluminum for ^{147}Pm. The ordinate of the bremsstrahlung spectrum is enlarged about 100 times.

7.4.3. Čerenkov radiation

The velocity of light in matter c depends on the refractive index n_r

$$c = \mathbf{c}\, n_r^{-1} \tag{7.17}$$

In water $n_r = 1.33$, in plexiglass 1.5. β-particles with energies >0.6 MeV move faster than light in water. When the particle velocity $(v_p) > c$, electromagnetic radiation is emitted coherently in a cone whose axis is the direction of the moving particle, Fig. 7.8(e). This is called Čerenkov radiation. The angle of the cone θ is obtained from

$$\sin\theta = c/v_p \tag{7.18}$$

Čerenkov radiation is continuous. Around the visible spectrum, the relative intensity per unit frequency is approximately proportional to the frequency. Thus, higher frequencies are more intense in Čerenkov radiation. This radiation is the source of the bluish light observed in highly radioactive solutions (Fig. 7.11) and around used reactor fuel elements submerged in water. The radiation can be used for detecting β-particles and for measuring high particle energies (from θ). For a fast electron the energy loss through radiation is ≤0.1% of the energy loss through other processes. Čerenkov detectors are described in Ch. 9.

7.4.4. Positron annihilation

Positrons interact with matter through ionization, excitation, emission of bremsstrahlung, and Čerenkov radiation in the same manner as negative electrons. As the kinetic

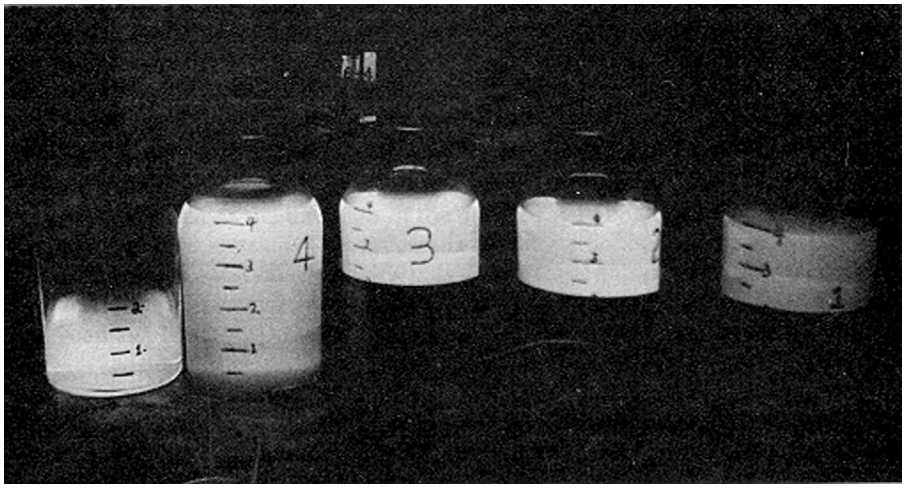


Figure 7.11 Bottles containing highly radioactive ^{90}Sr solutions glow in the dark due to Cerenkov radiation from daughter ^{90}Y (E_{max} 2.3 MeV).

energy of the positron decreases in the absorber, there is an increase in probability of direct interaction between the positron and an electron (Fig. 7.8(d)) in which both the positron and electron are annihilated. The energy of the two electron masses is converted into electromagnetic radiation. This process, known as *positron annihilation*, is a characteristic means of identification of positron emission. Since an electron mass is equivalent to 0.51 MeV, and the kinetic energy of the particles of annihilation is essentially zero, the total energy for the annihilation process is 1.02 MeV. In order to conserve momentum the photons must be emitted with equal energy and in exactly opposite direction in case of only two photons (the dominating case). These photons of 0.51 MeV each are referred to as *annihilation radiation*. The presence of γ-rays at 0.51 MeV in the electromagnetic spectrum of a radionuclide is strong evidence for the presence of positron emission by that nuclide. In rare cases three γ-rays of correspondingly lower energy are omitted.

7.4.5. Absorption curves and scattering of β-particles

An absorption curve for β-particles has a quite different shape than it has for α-particles (cf. Fig. 7.3). The continuous spectrum of energies in radioactive β-decay plus the extensive wide angle scattering of the β-particles by the absorber atoms account for the fact that range for β-particles continuously decrease. Even for a beam of initially mono-energetic electrons, the continuous removal of electrons from the beam path by wide angle deflection results in a plot showing a continuous decrease in the numbers of electrons with distance, with approximately 95% of the original β-particles stopped in the first half of the range. It is more common to speak of the

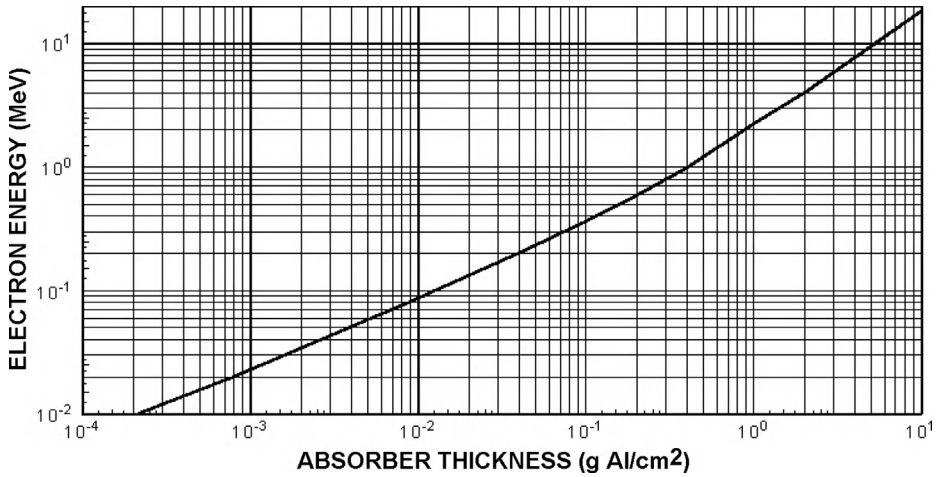


Figure 7.12 Empirical relation for the maximum range of β-particles in aluminum.

absorber thickness necessary to stop 50% of the particles than to speak of the range itself. This *half-thickness value* is much easier to ascertain experimentally than is an apparent range. It should be remembered that the energy deposited at complete β-absorption is $E_{abs} \approx 0.3\, E_{max}$ (Ch. 5).

The absorption curve for β-particles formed in radioactive decay can be described with fair approximation by Eqn (7.7). This is due to the continuous energy spectrum resulting in an exponential relationship for the range curve. In the E_{max} range 0.7—3 MeV the range in aluminum closely follows the relation (*Feather's rule*), with E_{max} in MeV:

$$\left(\text{g Al cm}^{-2}\right) = 0.543\ \ E_{max} - 0.160 \tag{7.19}$$

This is the range C_3 in Figure 7.4.

Figure 7.12 shows an empirical relationship between the maximum energy of β-particles and the extrapolated range in aluminum. Due to their relatively few interactions per travelled distance, compared to α-radiation, β-radiation has a much longer range. For example, the range of an α-particle of 5 MeV is 3.6 cm in air while that of a β-particle of 5 MeV is over 17 m. A comparison of the range in air and water for electrons and heavy particles is given in Table 7.2. Figure 7.13 is useful for a rapid estimate of the absorber thickness needed to protect against β-particles.

An additional complication in the experimental measurements of absorption curves for β-particles is found in the fact that a certain fraction of β-particles which are not originally emitted in the direction of the detector may be deflected to the detector by the large angle scattering. This process is known as *backscattering*, since the *backing* (or *support*) for radioactive samples may cause scattering of a certain fraction of the particles through as much as 180°. The fraction of back-scattered β-radiation depends on the geometry of

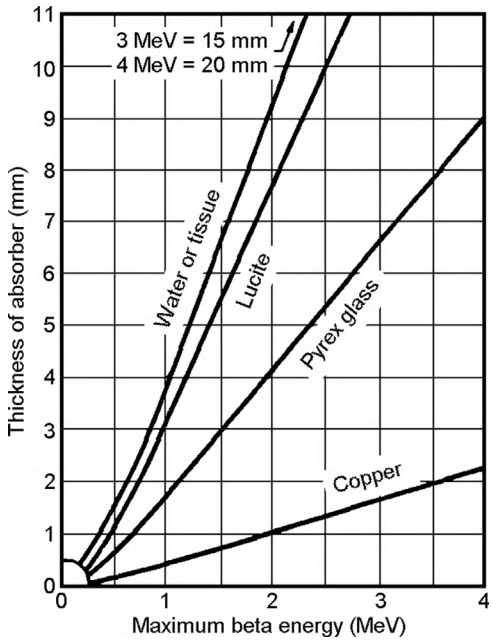


Figure 7.13 Thickness of various materials needed to completely stop β-particles.

the measuring system, the energy of the β-particles, and the thickness and electron density of the backing material. In Figure 7.14 the percent backscattering as a function of the atomic number of the backing material is shown for four β-energies (E_{max}); the radioactive sample is considered infinitely thin (i.e. no self-absorption). From the curve for beta rays from ^{32}P on platinum ($Z = 78$) we see that about 40% of the measured radiation is due to back-scattered radiation ($0.8/(1.0 + 0.8) = 0.4$). Backscattering increases with the thickness of the backing material up to a saturation value which is

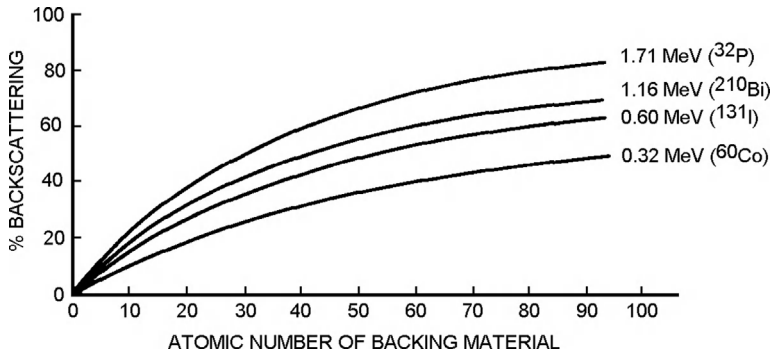


Figure 7.14 Backscattering of β-particles of different energy as function of the atomic number of thick backing materials.

reached when the thickness of the backing is about one-fifth of the extrapolated range of the β-particles in that material.

7.5. ABSORPTION OF γ-RADIATION

The absence of charge and rest mass for γ-rays results in little interaction with the absorbing atoms and in long ranges. The number of ion pairs produced in a given path length by γ-rays is only 1—10% of that produced by β-particles of the same energy (Fig. 7.7); e.g. a 1 MeV γ-ray produces only about one ion pair per centimeter of air. As a consequence of this low specific ionization of γ-rays, the ionization is almost completely secondary in nature resulting from the action of a few high energy primary ion pairs.

7.5.1. Attenuation coefficient

Unlike heavy particles and electrons which lose their energy as a result of many collisions, γ-rays are completely stopped in one or, at most, a few interactions. For thin absorbers the attenuation of γ-rays follows Eqn (7.8), where φ is the number of photons $m^{-2}\ s^{-1}$. The proportionality factor μ is called the (*total*) *attenuation coefficient*. When it has the dimension of m^{-1} and the thickness x is expressed in meters, μ is referred to as the *linear* attenuation coefficient. The attenuation coefficient can be expressed in other ways:

$$\mu_m = \mu/\rho = \sigma_a \mathbf{N}_A/M = \sigma_e Z\, \mathbf{N}_A/M \tag{7.20}$$

where ρ is the density, M the average atomic weight, and Z the average atomic number of the absorber. $\mathbf{N}_A\rho/M$ can be replaced by N_v, by which we can define a *macroscopic absorption cross-section* Σ (cf. §11.1). Σ^{-1} is the mean free path or *relaxation length* of the radiation in the absorbing material. μ_m (in $cm^2\ g^{-1}$ when x is in centimeters) is the *mass attenuation coefficient*; σ_a is the probability of reaction between a γ-ray and the electron cloud of the absorber atom (*atomic reaction cross-section*, m^2 atom^{-1}); σ_e is the probability of the reaction of a γ-ray with a single electron of the absorber (*electron reaction cross-section*, m^2 electron^{-1}). σ_a and σ_e are analogous to the nuclear reaction cross-sections discussed later (cf. Ch. 11). In Table 7.1 only the equivalent nuclear and atomic reaction cross-sections are given.

Since a γ-ray may be removed from the beam in the first few Ångströms of its entrance into the absorber or may travel several centimeters with no interaction at all and then be removed, it is not possible to apply the range concept to γ-rays in the way that it is applied to charged particles. However, it is experimentally easy to measure the thickness of absorber necessary to remove half of the initial γ-rays (half thickness value) from a beam, or reduce it to 1/10, 1/100, etc. Figure 7.15 shows the required thickness of concrete and lead necessary to reduce γ-rays of different energies by factors of 10 (shielding thicknesses are further discussed in §7.7). The half-thickness value is